

WASP-12b

R-0442

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_221202 · created 2026-07-19T14:26:54+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459915.8702 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.018
Transit depth (Rp/Rs)^2	1.79 +/- 0.49 [%]
Orbital Inclination (inc)	78.57 +/- 2.46
Airmass coefficient 1 (a1)	1.18 +/- 0.012
Airmass coefficient 2 (a2)	-0.155 +/- 0.059
Scatter in the residuals of the lightcurve fit is	1.0 %
Best Comparison Star	#3 - [451, 240]
Optimal Aperture	4.49
Optimal Annulus	9.26
Transit Duration (day)	0.1127 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.134 ± 0.018 vs 0.1178 (deviation 0.9σ)
Duration fit vs published	0.1127 d vs 0.125 d (deviation 1.27σ)
Mid-time O–C	1.8 min
Depth SNR	7.4
Residual scatter	1.0 %

Light curve

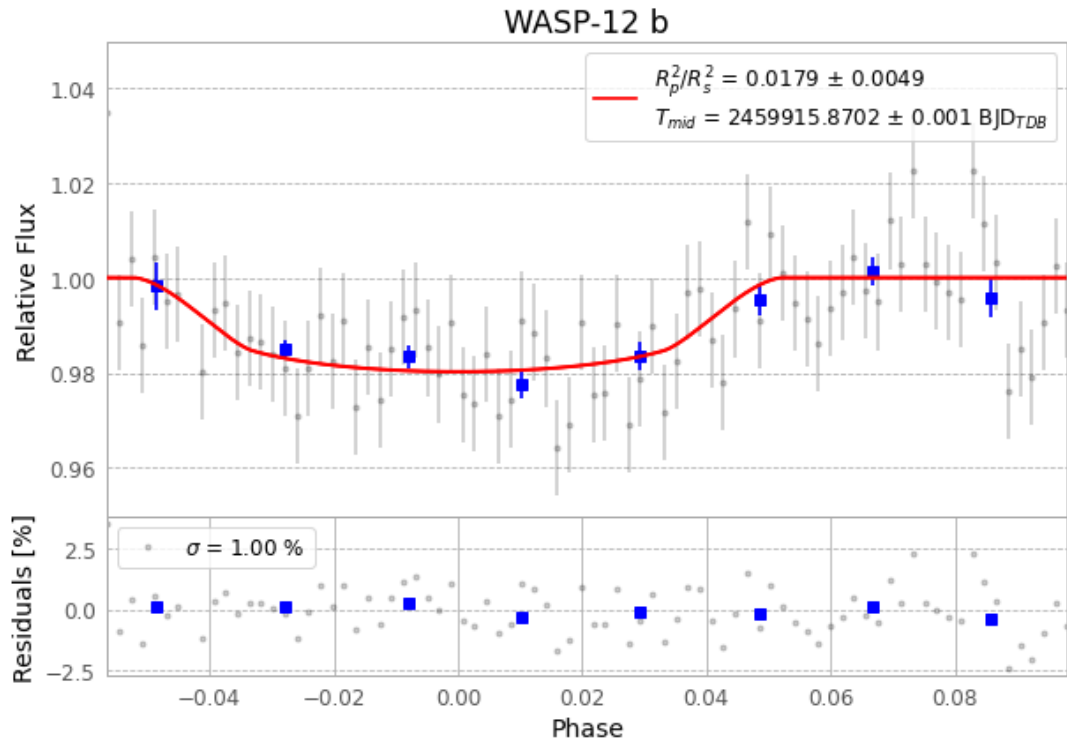


Figure 1 — FinalLightCurve_WASP-12 b_2022-12-02.png (SHA-256 c95d90d12d4e98db...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 161], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 32be4fce76373ba4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

82 FITS frames (54 MB), WASP-12221202071515.FITS → WASP-12221202111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-221202.log (1a80f176b56e...), FinalLightCurve_WASP-12 b_2022-12-02.png (c95d90d12d4e...), FinalLightCurve_WASP-12 b_2022-12-02.csv (678b9cab8b4e...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 14:26Z.